

# Matrices

## 4.2 Operations of Matrices

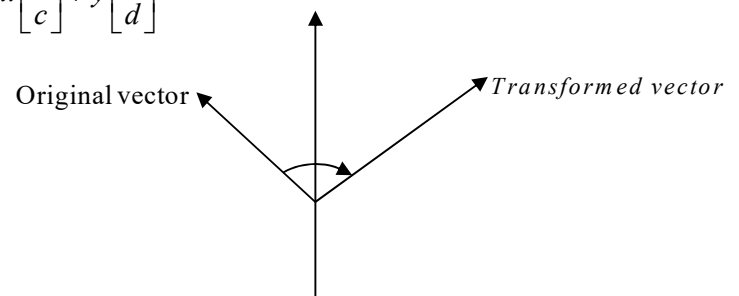
**(a) Addition and Subtraction**(Only between matrices of equal dimensions)

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2} \pm \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} (a \pm 1) & (b \pm 2) \\ (c \pm 3) & (d \pm 4) \end{bmatrix}$$

**(b) Multiplication**

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2} \begin{bmatrix} x \\ y \end{bmatrix}_{2 \times 1} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}_{2 \times 1} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix}$$

This can be seen as an example of Linear Transformation.



Here,  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  is the transforming matrix

being applied to a vector  $\begin{bmatrix} x \\ y \end{bmatrix}$ . The origin should be in the same position.

**Example:** Find the multiplication of matrices A and B.

$$(a) \quad A = \begin{bmatrix} 3 & 5 & -1 \\ 4 & 0 & 2 \\ -6 & -3 & 2 \end{bmatrix}_{3 \times 3} \quad \text{and} \quad B = \begin{bmatrix} 2 & -2 & 3 & 1 \\ 5 & 0 & 7 & 8 \\ 9 & -4 & 1 & 1 \end{bmatrix}_{3 \times 4}$$

$$\begin{aligned} \text{Solution: } AB &= \begin{bmatrix} 3 & 5 & -1 \\ 4 & 0 & 2 \\ -6 & -3 & 2 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 2 & -2 & 3 & 1 \\ 5 & 0 & 7 & 8 \\ 9 & -4 & 1 & 1 \end{bmatrix}_{3 \times 4} \\ &= \begin{bmatrix} 6+25-9 & -6+4 & 9+35-1 & 3+40-1 \\ 8+18 & -8-8 & 12+2 & 4+2 \\ -12-15+18 & 12-8 & -18-21+2 & -6-24+2 \end{bmatrix} \end{aligned}$$

$$AB = \begin{bmatrix} 22 & -2 & 43 & 42 \\ 26 & -16 & 14 & 6 \\ -9 & 4 & -37 & -28 \end{bmatrix}$$

$$(b) \quad A = \begin{bmatrix} 1 & 1 \\ 100 & 100 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\text{Solution: } AB = \begin{bmatrix} 1 & 1 \\ 100 & 100 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(c) \quad A = [3 \ 6 \ 1]_{1 \times 3} \quad \& \quad B = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}_{3 \times 1}$$

$$\text{Solution: } AB = [3 \ 6 \ 1]_{1 \times 3} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}_{3 \times 1} = [19]_{1 \times 1}$$

$$(d) \quad A = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}_{3 \times 1} \quad \text{and} \quad B = [3 \ 6 \ 1]_{1 \times 3}$$

$$\text{Solution: } AB = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}_{3 \times 1} \begin{bmatrix} 3 & 6 & 1 \end{bmatrix}_{1 \times 3} = \begin{bmatrix} 3 & 6 & 1 \\ 6 & 12 & 2 \\ 12 & 24 & 4 \end{bmatrix}_{3 \times 3}$$

$$(e) \ A = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 \\ 100 & 100 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 100 & 100 \end{bmatrix} = \begin{bmatrix} 99 & 99 \\ -101 & -101 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 1 \\ 100 & 100 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ -200 & 0 \end{bmatrix}$$

$$\Rightarrow AB \neq BA$$